

BC 11. Let $y = x\sqrt{1+x^2}$. When $x = 0$ and $dx = 2$, the value of dy is

1973
NC

- (A) -2 (B) -1 (C) 0 (D) 1 (E) 2

BC 14. If $y = x^2 + 2$ and $u = 2x - 1$, then $\frac{dy}{du} =$

1969
NC

- (A) $\frac{2x^2 - 2x + 4}{(2x - 1)^2}$ (B) $6x^2 - 2x + 4$ (C) x^2
(D) x (E) $\frac{1}{x}$

BC 36. The approximate value of $y = \sqrt{4 + \sin x}$ at $x = 0.12$, obtained from the tangent to the graph at $x = 0$, is

1969
NC

- (A) 2.00 (B) 2.03 (C) 2.06 (D) 2.12 (E) 2.24

BC 9. If $y = \cos^2 3x$, then $\frac{dy}{dx} =$

1973
NC

- (A) $-6 \sin 3x \cos 3x$ (B) $-2 \cos 3x$ (C) $2 \cos 3x$
(D) $6 \cos 3x$ (E) $2 \sin 3x \cos 3x$

3. The wind chill is the temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), a human feels based on the air temperature, in degrees Fahrenheit, and the wind velocity v , in miles per hour (mph). If the air temperature is 32°F , then the wind chill is given by $W(v) = 55.6 - 22.1v^{0.16}$ and is valid for $5 \leq v \leq 60$.

2007B
BC 6C

- (a) Find $W'(20)$. Using correct units, explain the meaning of $W'(20)$ in terms of the wind chill.
(b) Find the average rate of change of W over the interval $5 \leq v \leq 60$. Find the value of v at which the instantaneous rate of change of W is equal to the average rate of change of W over the interval $5 \leq v \leq 60$.
 (c) Over the time interval $0 \leq t \leq 4$ hours, the air temperature is a constant 32°F . At time $t = 0$, the wind velocity is $v = 20$ mph. If the wind velocity increases at a constant rate of 5 mph per hour, what is the rate of change of the wind chill with respect to time at $t = 3$ hours? Indicate units of measure.